



Digitalization of Mess Waste Data Collection

The aim is to digitalize the collection and analysis of hostel mess waste data at IIT Bombay to streamline waste data collection, enhance data accuracy, and enable effective visualization for informed decision-making.

Technology Stack

The solution leverages the Microsoft ecosystem for automation, data storage, and visualization:

- **Power Apps** – For data entry and user interface.
- **SharePoint List** – For structured data storage.
- **Power Automate** – For workflow automation and approval processes.
- **Power BI** – For data visualization and analysis.

System Implementation

1. Power Apps Setup

Power Apps is used to build a data collection form that allows hostel mess staff to enter waste data.

Steps to Set Up Power Apps:

1. **Create a SharePoint Site:**
 - A dedicated SharePoint library is created.
 - The current user is added as a member.
2. **Create a SharePoint List:**
 - Fields required: Date, Hostel Name, Vegetable Waste (kg), Mess Surplus (kg), Leftover Food (kg), Total Waste (kg), Per Capita Waste.
 - Some fields use formulas for automatic calculations.



3. Form Development in Power Apps:

- Use **Integrate Option** from the SharePoint list to build a Power App.
- Add input fields matching the SharePoint List structure.
- Add a **Submit Button** with validation to ensure all fields are filled before submission.
- Display a **Success Message** upon submission.
- **Fix the Hostel Field as Disabled** (DisplayMode.Disabled) to prevent users from changing it.
- **Set Default Date to Today()**.
- Use **Conditional Logic** (if-else) for automatic calculations (e.g., Per Capita Waste based on hostel size) in the sharepoint list itself.

Mess Waste Tracker

Hostel

H2

Date

1/8/2025

Leftover Food (Kg)

Mess Surplus (Kg)

Vegetable Waste (Kg)

Total Waste (Kg)

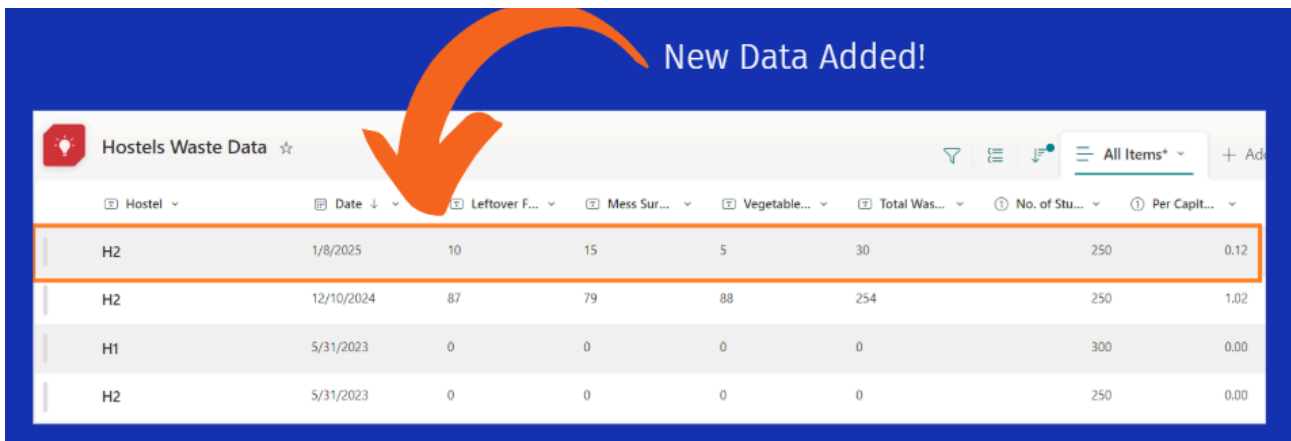
0

Submit Data



2. SharePoint List for Data Storage

- Acts as the central database for storing waste data.
- Automatic calculations enabled:
 - **Per Capita Waste Calculation** using student numbers from each hostel.
 - **Total Waste Calculation** based on input values.
 - **Verification Status Column**: An additional column is included to store the verification status of each entry. Whenever a new entry is submitted through the Power App, a request is automatically generated via Power Automate, and the status is updated in this column accordingly.
- Data is structured for smooth integration with Power BI.



New Data Added!

Hostel	Date	Leftover F...	Mess Sur...	Vegetable...	Total Was...	No. of Stu...	Per Capit...
H2	1/8/2025	10	15	5	30	250	0.12
H2	12/10/2024	87	79	88	254	250	1.02
H1	5/31/2023	0	0	0	0	300	0.00
H2	5/31/2023	0	0	0	0	250	0.00

3. Power Automate for Approval Workflow

To validate and approve data collected via Power Apps, **Power Automate** automates the workflow.

Steps to Configure Power Automate:

1. Open **Power Automate** in Microsoft 365.
2. Click **Create** → **Automated Flow** and select a name.
3. Select **Trigger**: "When an item is created" in SharePoint List.
4. Search for and add "Start and wait for an approval".



5. Configure approval rules to validate entries before analysis.
6. **Automatically trigger approval workflow:** Whenever a new entry is submitted through Power Apps, Power Automate will automatically generate an approval request and update the verification status column in the SharePoint List.

Power Automate

Please review the Hostel Waste Data



Requested by Madhav Agrawal <23b1539@iitb.ac.in>

Data added for Hostel : H2, Leftover Food : 10, Mess Surplus : 15, Vegetable Waste: 13, Total Waste : 38

[To view the detailed data, click here](#)

Date Created: Friday, January 10, 2025 5:05 AM GMT

Approve >

Reject >

View this Approval on the Power Automate Portal [here](#)



Verification Status Updated!

All Items*		
① Per Capit...		Verificati...
50	0.12	Approved
50	1.02	
00	0.00	Approved
50	0.00	Approved



4. Data Visualization in Power BI

Power BI is used to create interactive dashboards for trend analysis and comparison.

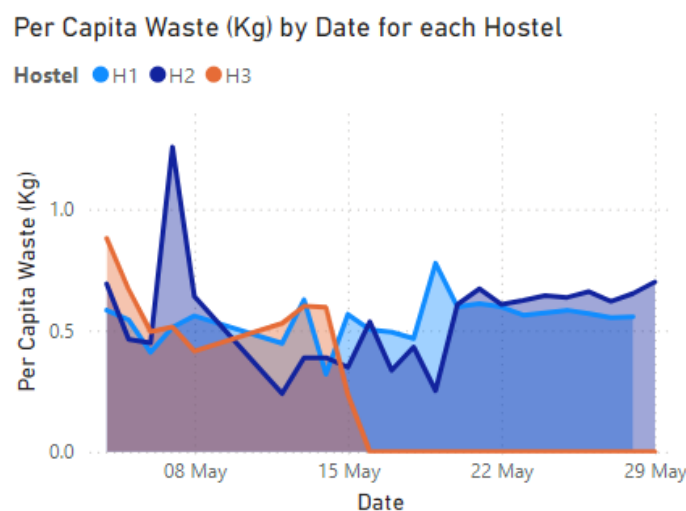
Steps to Implement Power BI Visualizations:

1. Power BI Setup:

- Enter the **SharePoint site address** to connect Power BI to the dataset.
- Load the **SharePoint List data** into Power BI.
- Use the **Transform Data** option to clean and restructure the dataset:
 - Correct column names for clarity.
 - Change column types (e.g., numerical values to integers/decimals, date column to date format).
- Ensure proper **data type assignments** to enable accurate calculations and analysis.

2. Creating Visualizations:

- **Table and Line Charts:** Start building visual representations such as tables, line charts, and summary cards.
- **Filters:** Apply filters to select specific hostel data (e.g., only for H1).
- **Formatting:**
 - Change axis titles, labels, and legends via the **Format Visual** option and rename visuals for better readability.
- **Line Graph for Total Waste:**
 - Use **Hostel** as a legend to display individual trends per hostel.
- **Donut Chart:** Represent the composition of various waste types across hostels.
- **Bar Graph:** Compare waste generation trends across hostels.





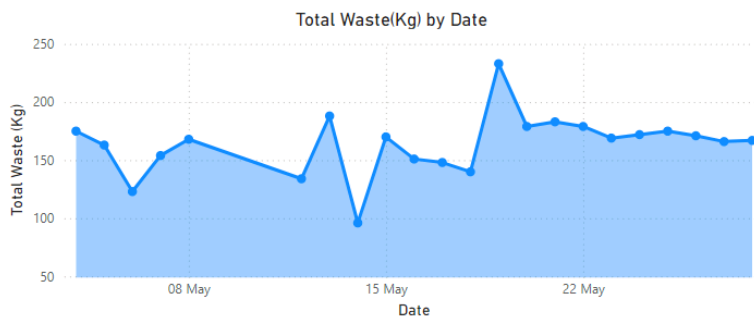
Hostel	Date	Vegetable Waste(Kg)	Mess Surplus(Kg)	Leftover Food(Kg)	Total Waste(Kg)	Per Capita Waste
H1	28 May 2022	26	37	104	167	
H1	27 May 2022	25	38	103	166	
H1	26 May 2022	24	39	108	171	
H1	25 May 2022	22	43	110	175	
H1	24 May 2022	19	42	111	172	
H1	23 May 2022	18	45	106	169	
H1	22 May 2022	13	62	104	179	
H1	21 May 2022	15	59	109	183	
H1	20 May 2022	13	62	104	179	
H1	19 May 2022	23	90	120	233	
H1	18 May 2022	30	68	42	140	
H1	17 May 2022	42	49	57	148	
H1	16 May 2022	35	46	70	151	
H1	15 May 2022	21	68	81	170	
H1	14 May 2022	15	29	52	96	

0.55

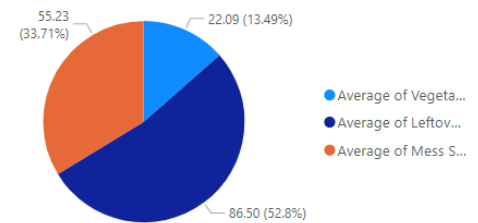
Average of Per Capita Waste (Kg) of H1

163.82

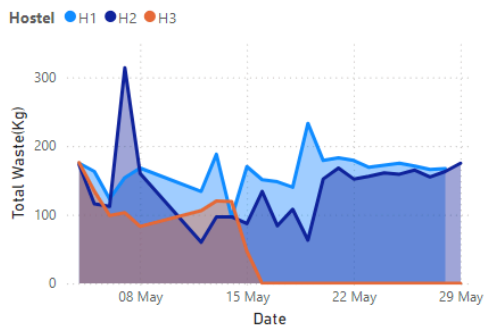
Average of Total Waste(Kg) of H1



Comparison of Wastes

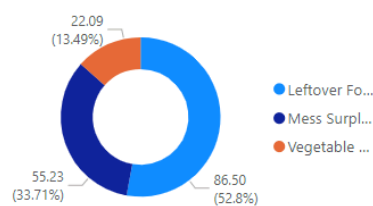


Total Waste (Kg) by Date for each Hostel

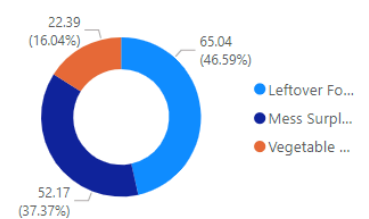


Mess Surplus Vs Vegetable Waste Vs Leftover Food

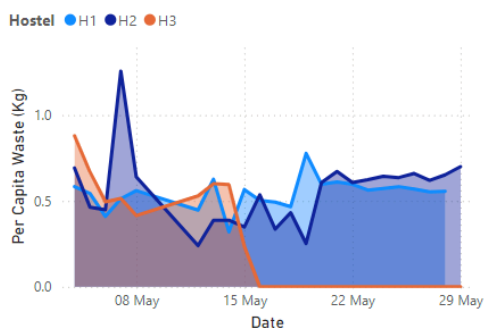
Hostel 1



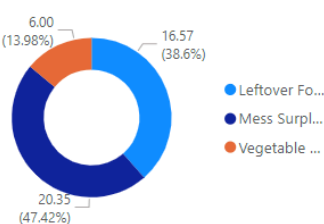
Hostel 2



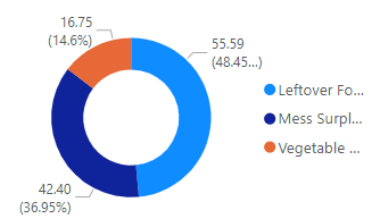
Per Capita Waste (Kg) by Date for each Hostel



Hostel 3



All Hostels





Limitations

Current Challenges with Power Apps

- The Power Apps solution requires users to have an **IIT Bombay LDAP account**, which means mess workers or managers who do not have LDAP access cannot use the application outside the IITB environment.
- This limits accessibility and prevents external or non-technical users from directly entering data into the system.

Proposed Solution & Ongoing Work

- To address this issue, we are working on developing a **Sustainability Cell Portal**, where mess workers can enter data directly via a web app.
- This portal will store data in **Google Sheets or SharePoint Lists**, eliminating the dependency on LDAP.
- However, the **pipeline is still under development**, and there are technical challenges that need to be resolved for seamless integration.

 **WasteTracker**

Name:

Date of entry:

Location:

☐ Academic Area

☐ Staff Area

☒ Hostel Area

Hostel No.:

NEXT →



Future Improvements

To ensure data accuracy and scalability, the following recommendations are proposed:

1. **Automated Data Entry Enhancements:**
 - Integrate IoT-based waste measurement sensors.
2. **Advanced Data Analytics:**
 - AI-based anomaly detection for identifying data inconsistencies.
 - Predictive analytics to forecast waste generation trends.
3. **User Access & Roles:**
 - Assign specific permissions to hostel managers, mess supervisors, and students.
4. **Alerts:**
 - Enable push notifications for alerts and approvals.

References

Below are the relevant links to the app and lists:

1. **Power App:** [Mess Waste Tracker - Power App](#)
2. **Sharepoint Site:** [WasteManagement - SharePoint](#)
3. **SharePoint List:** [Hostels Waste Data - SharePoint List](#)